

Faculty of Nursing

Course Title: Parasitology

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Assignment: common parasitic diseases in Egypt

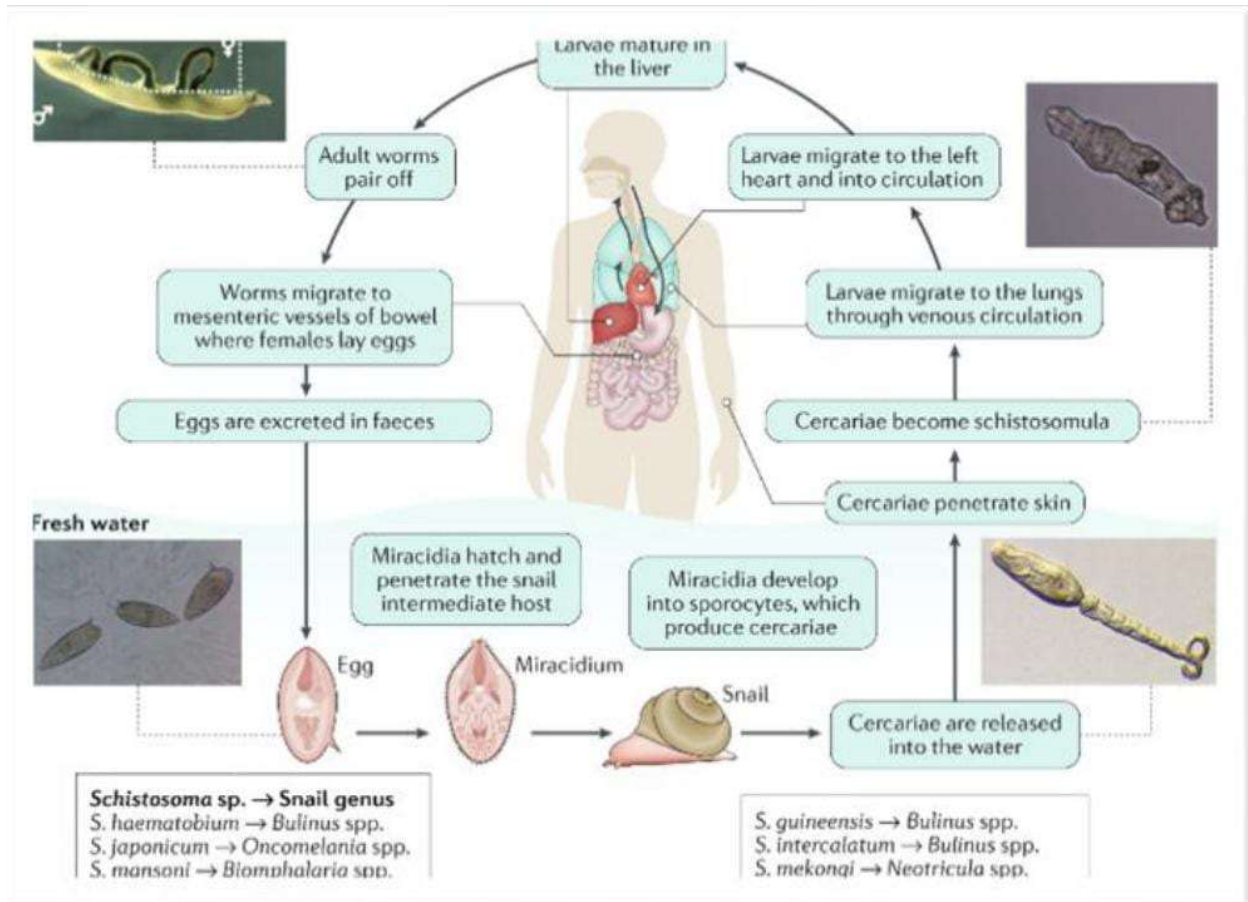
Common parasitic diseases in Egypt

Schistosomiasis is a parasitic disease caused by blood flukes (Trematodes) of the genus *Schistosoma* (S.). It is well documented that schistosomiasis haematobium was endemic in Ancient Egypt. Infection was diagnosed in mummies 3000, 4000 and 5000 years old. Scott was the first to describe the pattern of schistosomiasis infection in Egypt.

Schistosomiasis haematobium was highly prevalent (60%) both in the Nile Delta and Nile Valley South of Cairo in districts of perennial irrigation while it was low (6%) in districts of basin irrigation. *Schistosoma mansoni* infected 60% of the population in the Northern and Eastern parts of the Nile Delta and only 6% in the Southern part. Neither *S. mansoni* cases nor its snail intermediate host were found in the Nile Valley South

of Cairo. The building of the Aswan High Dam -which was completed in 1967 – did not cause any increase in schistosomiasis prevalence. In 1990, a study conducted in nine governorates of Egypt confirmed the change in the pattern of schistosomiasis transmission in the Delta. There was an overall reduction in *S. mansoni* prevalence while *Schistosoma haematobium* had continued to disappear. In Middle and Upper Egypt there was consistent reduction in the prevalence of *S. haematobium* except in Sohag, Qena, and Aswan governorates. However, foci of *S. mansoni* were detected in Giza, Fayoum, Menya and Assiut. All schistosomiasis control projects implemented in Egypt from 1953 to 1985 adopted the strategy of transmission control and were based mainly on snail control supplemented by anti-bilharzial chemotherapy. In 1997, the National Schistosomiasis Control Program (NSCP) was launched in the Nile Delta. It adopted morbidity control strategy with Praziquantel mass treatment as the main component. In 1996, before the NSCP, 168 villages had *S. mansoni* prevalence >30%, 324 villages 20–30% and 654 villages 10–20%. By the end of 2010, in

the whole country only 29 villages had prevalence >3% and none had more than 10%.



Intestinal parasites (IPs) are widespread throughout the world and have a significant negative influence on morbidity and mortality, particularly in developing countries^{1,2}. IPs have been linked to stunting, physical weakness, and poor academic performance in children. By interacting with absorptive surfaces, physically

obstructing the intestinal lumen, producing proteolytic chemicals, and consuming nutrients destined for the body, IPs reduce nutrient intake in the body. Due to their heightened susceptibility to nutritional inadequacies, children are more vulnerable to the effects of intestinal parasitic infections^{3,4}.

Furthermore, IPs-related disorders display a heavy burden. Over two billion people are affected globally by IPs, of which 300 million have severe morbidity and approximately 4.5 billion people are at risk. The World Health Organization reported that there were 200 million instances of *Giardia lamblia*, 800 million cases of *Ascaris lumbricoides*, 700 million cases of hookworm, 500 million cases of *Trichuris trichuria*, and 500 million cases of *Entamoeba histolytica*/ *dispar* worldwide.

According to earlier estimates, IPs result in an annual loss of 39 million disability-adjusted life years, resulting in serious health and financial problems. However, the biggest problem of IPs is that 90% of infected people are asymptomatic^{5,6}.

The main intestinal protozoans that infect humans are *Entamoeba histolytica*, *Balantidium coli*, *Giardia lamblia*, *Isospora belli*, and *Cryptosporidium* species⁷. Other species, like *Blastocystis* sp.

and *Dientamoeba fragilis*, are disputed pathogens, their presence is not associated with clinical significance and their role in people who have digestive symptoms is currently debatable. However, few clinics and laboratories routinely test for *Dientamoeba fragilis* and *Blastocystis* sp.⁸.

Similarly, many other non-pathogenic intestinal protozoa can be found in humans that are not associated with diseases, including *Entamoeba coli*, *Entamoeba dispar*, *Entamoeba hartmanni*, *Entamoeba polecki*, *Endolimax nana*, *Iodamoeba bütschlii*, and *Chilomastix mesnili*⁹.

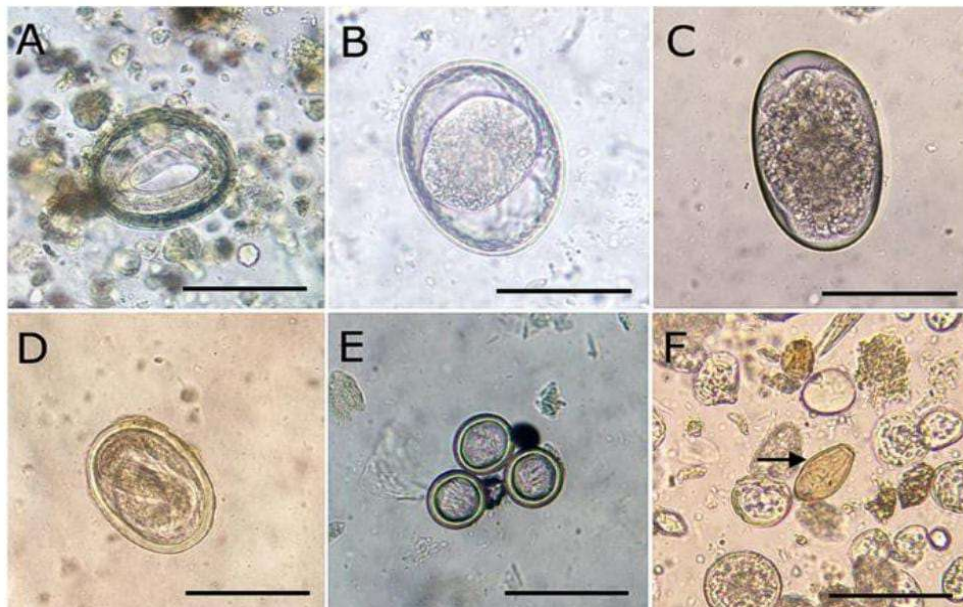
Intestinal helminths including *Ascaris lumbricoides* (*A. lumbricoides*), *Trichuris trichiura* (*T. trichiura*), and hookworm species (*Necator americanus* and *Ancylostoma duodenale*) are grouped under

soiltransmitted helminths, which are highly prevalent in developing countries¹⁰. In Egypt, intestinal schistosomiasis (*S. mansoni*) is prevalent in the Nile Delta while urogenital schistosomiasis (*S. haematobium*) in the Nile valley south of Cairo, down to Aswan and beyond. Epidemiological studies have shown that in endemic areas, multiparasitism is more frequent than single infections^{11,12,13}. Geographical, behavioral, biological, and socioeconomic factors all influence how prevalent these illnesses are. IPs are directly related to the hot and humid climate of the tropics, the lack of access to clean water, the unsanitary environmental conditions, overcrowding, and low family income. These circumstances facilitate intestinal parasite growth, transmission, and exposure¹⁴. The majority of research on co-infections with several parasites, which occurs when infectious agents coexist in the same host and can have a prevalence of up to 80% in some populations, has focused on humans¹⁵. These co-infections may result from a variety of environmental factors, such as contagious settings and transmission paths, and may be

aided or hindered by direct or indirect interactions between different parasite species¹⁶. The interaction between different parasite populations (within and between hosts) and host populations, rather than associations play a significant role¹⁷. Though the specifics of these interactions within the host are unknown, some are sought to be synergistic where the presence of one parasite may encourage the occurrence of future parasitic infections, while others are thought to be antagonistic where parasites compete for the same ecological niche within the host¹⁸.

Studies carried out in several governorates over the past few years to determine the prevalence of IPs and the risk factors associated with them among the Egyptian population have found that IPs continue to be a problem for public health. Therefore, for adequate planning and execution of efficient preventative and control strategies, frequent assessment of IPs prevalence and identification of the related risk factors is required. Therefore, the current study's objective was to investigate the prevalence of multiparasitism and intestinal parasites. Additionally, to study the

relationships between multiple parasite infections and their risk factors in comparison to single parasite infections or complete lack of infection in the Egyptian population.



Cutaneous leishmaniasis (CL) is a neglected clinical form that is highly prevalent in the northern and eastern parts of Egypt. *Leishmania major* and *Leishmania tropica* circulate in Sinai (Shehata et al. 2009). *L. major* is known to be transmitted by the sandfly *Phlebotomus papatasi*; however, no vector information is available in the sole report documenting *L. tropica* from Egypt (Shehata et al. 2009). A new focus of CL was identified

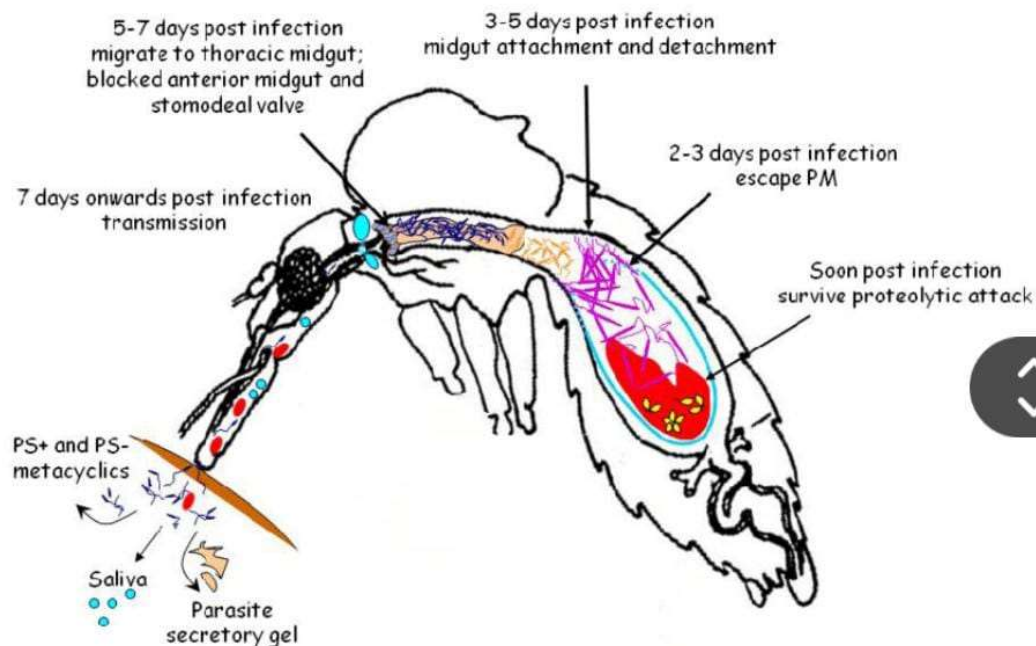
on the border between Egypt and neighbouring Palestine and Israel (Jacobson 2003, Nasereddin et al. 2008). The presence of CL caused by *L. tropica* remains unclear with regard to the two scenarios described by Shehata et al. (2009): that historical difficulties in distinguishing *L. major* from *L. tropica* (Jacobson 2003) have masked the presence of *L. tropica* in past studies vs. that recent incursion by *L. tropica* from adjacent endemic regions is responsible for this species' recent detection.

Leishmaniasis diagnosis is a major challenge due to its diverse clinical manifestations, which make the concrete diagnosis of present and past cases difficult. However, differential diagnosis is important because other diseases with a similar clinical spectrum [e.g., leprosy, skin cancer and tuberculosis for CL, malaria and schistosomiasis for visceral leishmaniasis (VL)] often co-occur in areas of endemicity (Reithinger et al. 2007). The ability to distinguish between *Leishmania* species is, therefore, crucial for correct diagnosis, prognosis, treatment and control measures. In Egypt, however, little is known about parasite genotypes

(Shehata et al. 1988). Recent studies have depended on internal transcribed spacer (ITS)-1 sequences to differentiate between imported and autochthonous infections (Jacobson et al. 2003); other methods can be used to complement ITS-1 sequences, such as microsatellites and kinetoplastid DNA (Schwenkenbecher et al. 2006). Epidemics of leishmaniasis are known to occur in many foci throughout the world (Seaman et al. 1996, Rodríguez-Barraquer et al. 2008, Wang et al. 2010).

Nonetheless, molecular studies to date have not often taken advantage of phylogenetic or population genetic approaches for data analysis in Middle Eastern countries (Tibayrenc 2005, Nasereddin et al. 2008), even though successful examples are available from other locations (Katakura 2009, Miranda et al. 2009). Consequently, CL public health impacts have been underestimated, as a substantial number of cases go unrecorded: approximately 1.5-2 million new cases are estimated to occur annually, but only 600,000 are officially declared (WHO 2008).

Leishmaniasis dynamics have changed in response to environmental, demographic and human behavioural factors (Campbell-Lendrum et al. 2001). The response of leishmaniasis to environmental changes may result from a change in the geographic distribution of the potential vectors/animal reservoir, either by the identification of new habitats or by range shifts for the same species (Parmesan 2006). In Egypt, however, CL cases are still underestimated due to the Bedouin traditions of preventing females from visiting clinics and their dependence on routine heat therapy for the treatment of CL (Samy 2009). Here, we report our findings from a comprehensive study conducted over seven years to track the CL transmission cycle. Our findings suggest the continuous circulation of *L. major* in Sinai and identify both the vector(s) and animal reservoir(s), providing new insight into the ecology of CL and disease transmission in Egypt.



Lymphatic filariasis

Lymphatic filariasis has been eliminated from most areas of Egypt. However, sustained mass drug administration activities and post-mass drug administration surveillance are still required. The mapping of chronic infected villages has been completed and they are being targeted by programme activities.

WHO works with the MoHP building the capacity to carry out epidemiological surveillance, which is essential for the monitoring and evaluation of disease

occurrence and impact of mass drug administration activities. Further effort needs to be placed on alleviating the suffering of chronic lymphatic filariasis patients. With increased political commitment and sustained programme efforts the elimination goal can be achieved.

